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ÿ Overview

The OCP8195X is a non-isolated step-down LED constant current driver actuator, suitable for 85VAC-265VAC full range input voltage applications use. The chip adopts constant peak turn-off, inductor current critical turn-on Operating mode.

OCP8195X integrates high-voltage JFET start-up and power supply circuit, speeding up system startup and reducing external components, including eliminating Vcc capacitance.

The OCP8195X integrates various protection mechanisms. Chip integration patent The power tube overload protection function, over temperature drop current function, night light, Enable control and PWM dimming and short circuit protection functions. System short circuit , the chip operates at 2K frequency.

The chip can work in the normally-on state, which is convenient for mass production. Control of MOS power capability. Coupled with overload protection function, you can Effectively prevent the problem of bombing caused by differences in power devices.

The chip is packaged in DIP-7L/SOP7L, the operating temperature range range from -40°C to 85°C.

ÿ Features

ÿ 85VAC to 265VAC input voltage range
ÿ High voltage JFET startup and power supply
ÿ Ultra-fast LED startup time
ÿ Ultra-low working current
ÿ Excellent line/load regulation
ÿ Critical conduction mode for high efficiency
ÿ Built-in 500V power MOSFET
ÿ Output short circuit protection
ÿ Overload and overtemperature protection
ÿ Cycle-by-cycle overcurrent limit
ÿ RoHS green material package
ÿ Support PWM, dual color temperature dimming
ÿ No Vcc capacitor required
ÿ -40ÿ to +85ÿ ambient operating temperature

ÿ Application

ÿ Common LED lighting

ÿ Typical application circuit

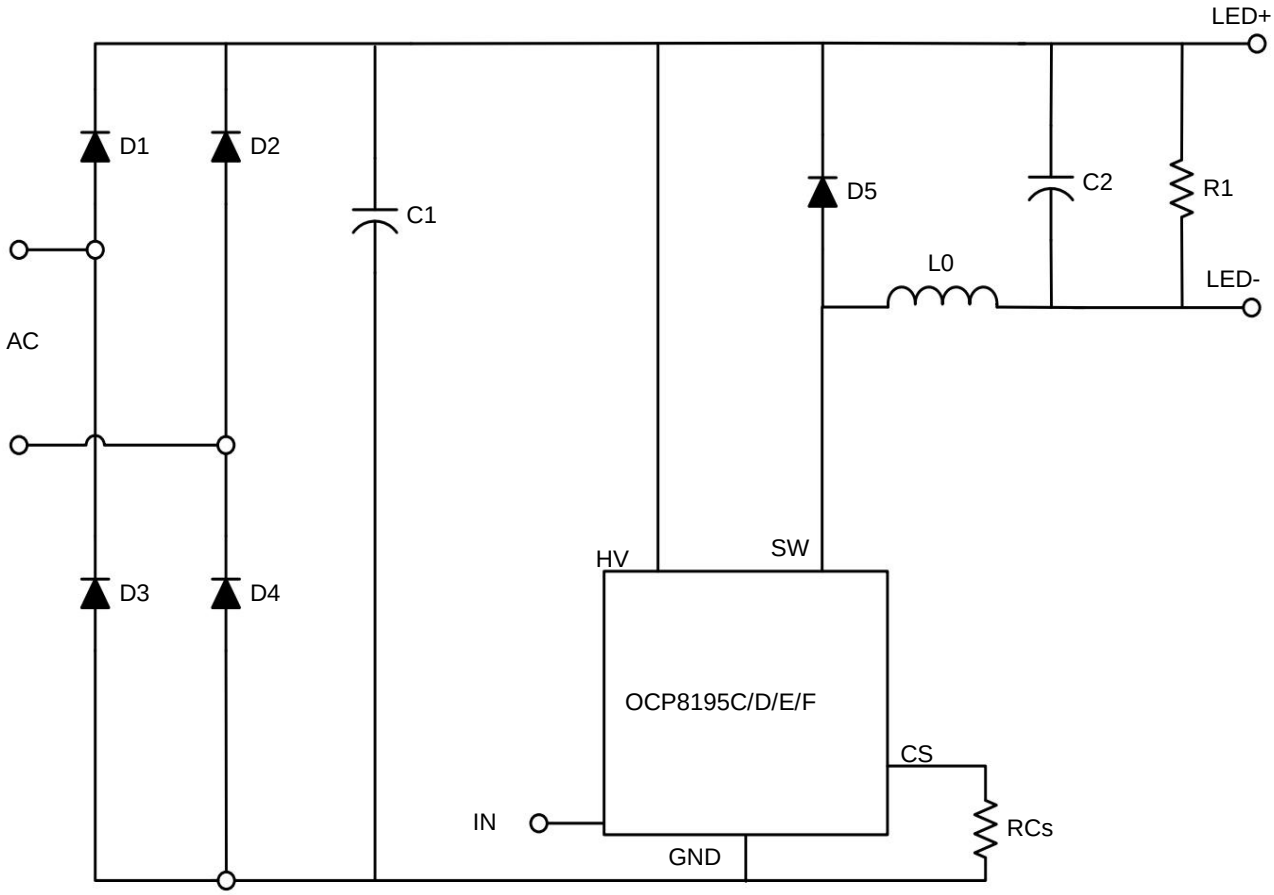
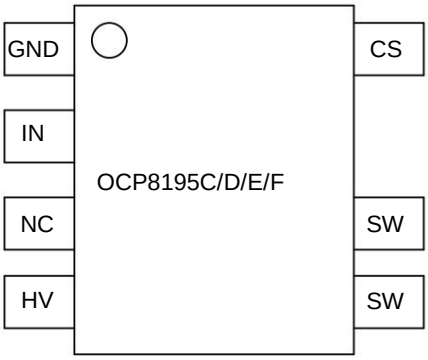


Figure 1 Typical application circuit of OCP8195X

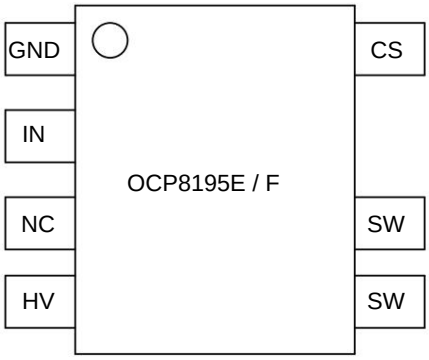
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ÿ Pin Definition



SOP-7L



DIP-7L

ÿ Pin description

pin name	pin number		describe
	SOP-7L	DIP-7L	
SW	5/6	5/6	Drain terminal of internal 500V MOSFET high voltage tube
CS	8	8	Current sampling terminal
NC	3	3	not connected
IN	2	2	Chip enable terminal, can be floated, can be connected to PWM signal for dimming
GND	1	1	chip ground
HV	4	4	High voltage terminal, connect to rectifier power supply

ÿ Circuit block diagram

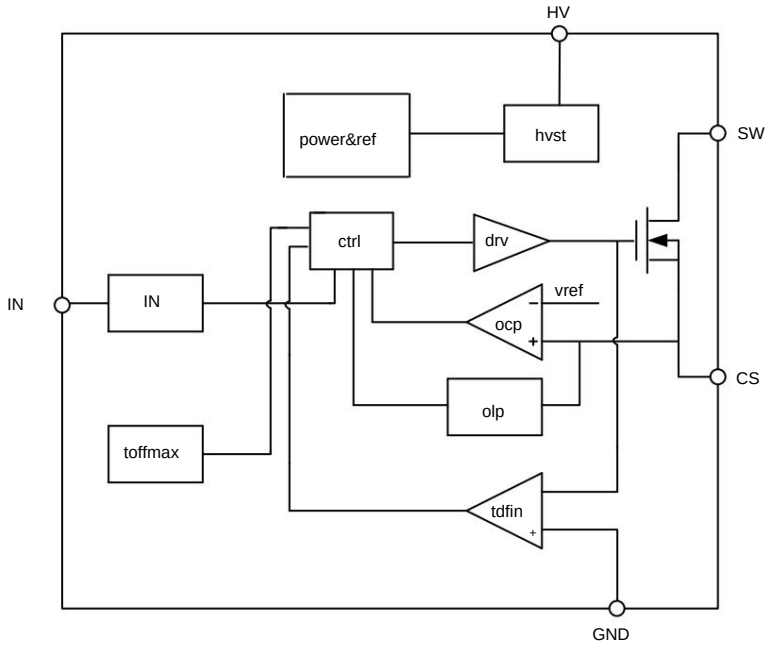


Figure 2 Internal module circuit diagram

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ÿ Absolute Maximum Ratings (Note 1)

parameter	symbol	scope	unit
SW ÿ HV	HV, SW pin to GND pin	-0.3 to +500.0	IN
ENÿCS	EN, CS pin to GND pin	-0.3 to +7.5	IN
TS	Storage temperature range	-55 to +150	°C
TJ	Operating Junction Temperature Range	-40 to +150	°C
TLEAD	Maximum Soldering Temperature (on pins, 10 seconds)	300	°C

ÿ Recommended operating temperature range (Note 2)

symbol	parameter	scope	unit
PER	Working temperature	ÿ40 ~ 85	°C

ÿ Recommended application range (Note 3)

Product model (package)	Application conditions		Maximum operating current	unit
	Input voltage	The output voltage		
OCP8195C	85V~265V	72V	160	mA
OCP8195D	85V~265V	72V	200	mA
OCP8195E(SOP7)	85V~265V	72V	280	mA
OCP8195E(DIP7)	85V~265V	72V	330	mA
OCP8195F(SOP7)	85V~265V	72V	330	mA
OCP8195F(DIP7)	85V~265V	72V	380	mA

Note 1: The maximum limit value means that beyond the working range, the chip may be damaged.

Note 2: Beyond the recommended operating temperature range, the chip cannot be guaranteed to work normally.

Note 3: Beyond the recommended application range, there is no guarantee that the chip will not reduce the current before the over-temperature point of 145°C.

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ÿElectrical parameters (test conditions: if there is no special instructions, TA = 25ÿ)

symbol	parameter	condition	Min	Typ	Max	Units
Power supply voltage (HV **Pin)**						
Iq	Operating current (without switch)		·	120		uA
Vst	Starting voltage	HV rise	·	12.3	·	IN
Vuvlo	Undervoltage protection voltage	HV drop	·	7.5	·	IN
time						
Toffmax	Maximum off time		·	540	·	us
Tlebon	Leading edge blanking time		·	500	·	ns
Tleboff	Turn off blanking time		·	2	·	us
Current Sampling (CS **Pin)**						
you	CS pin current limit		·	600	·	mV
Vcr	Normally On Current Reference		·	200	·	mV
EN threshold						
ENH	Chip On Threshold		·	1.38	1.7	IN
ENL	Chip Shutdown Threshold		1	1.28	·	IN
R	Pull-up resistor on EN pin		·	125	·	kÿ
Overheating protection						
TSD	Superheat regulation temperature point		·	145	·	ÿ
Power MOSFET (SWPin)						
VBR	Drain-source breakdown voltage of power tube		500	·	·	IN
Rdson	Static drain-source resistance	OCP8195C	·	12	·	ÿ
		OCP8195D	·	9	·	
		OCP8195E	·	4.8	·	
		OCP8195F	·	2.2	·	

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ÿ Application information

The OCP8195X contains a high voltage power MOSFET switch, high voltage JFET and a power control chip. The integrated high-precision constant current control and high-reliability protection function make it suitable for LED lighting applications.

1. Power supply

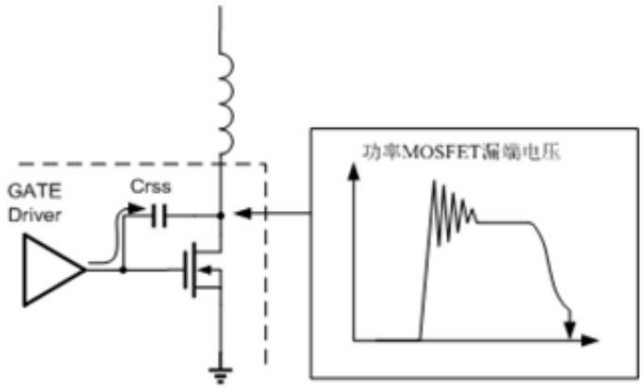
After the system is powered on, the chip directly supplies power from the bus through the high-voltage JFET. When the bus voltage reaches the chip turn-on threshold, the internal control circuit of the chip starts to work. When the chip is working normally, the required working current is still provided to it through the internal JFET.

2. Very low operating current

The OCP8195X operates at 120uA (typ), and the extremely low operating current contributes to higher efficiency.

3. Current zero-crossing detection without auxiliary winding

In the OCP8195X, the detection of the current zero-crossing point is realized by monitoring the coupling current of the gate-drain parasitic capacitance Crss of the MOSFET. When the inductor current freewheels to After zero, the variation of the drain voltage is controlled by the resonance of the inductance and the MOSFET drain capacitance. The voltage oscillation on the parasitic capacitance Crss will cause the current to change, and the leakage current will A negative current is generated when the voltage oscillation decreases, and a positive current is generated when the drain voltage oscillation increases. By detecting the change of this current to determine the zero-crossing point of the inductor current, As shown below:



Current zero-crossing detection

4. Quasi-resonant step-down (QR-Buck) constant current drive

In quasi-resonant buck mode, the chip keeps the CS peak current constant and starts a new PWM cycle at the first resonant valley. Therefore, it can be Realize high-precision constant current control and high conversion efficiency. The average LED regulator output current is given by:

$$I_{OUT}(mA) \approx \frac{1}{2} \cdot \frac{600\,mV}{RCs(\Omega)} \quad (1)$$

In the above formula, RCS is the sampling resistor connected between the CS pin and the ground of the step-down conversion system.

5. Line voltage compensation and leading edge blanking

When the MOS is turned on, the leading edge blanking circuit is integrated inside the chip. During the leading edge blanking time (typical value is 500ns), the internal circuit samples the cs voltage for the line voltage Compensation, in addition the cycle-by-cycle current detection comparator is invalid, so the MOS will be forced to turn on. After the leading edge blanking time expires, the peak detection comparator detects the peak normally current to control the power MOS off. The line voltage compensation circuit comes into play, using the cs voltage detected during the blanking time to compensate for the current peak, thereby eliminating the line

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Peak variance due to voltage and turn-off delay for good line regulation.

6. Protection

OCP8195X has built-in multiple protection functions, including LED short-circuit protection, HV under-voltage protection, chip temperature overheat regulation overload protection, etc.

When the LED is short-circuited, the chip works in the maximum off-time state and enters the low-frequency operating mode, which reduces system power consumption and prevents peak current overshoot, thereby protecting The system will not be damaged.

When the HV voltage is lower than 7.5V, the chip is turned off, and the power MOS is controlled to be normally off to prevent damage due to insufficient MOS drive capability.

Overload protection, if the system temperature is high and the output current setting is large, exceeding the MOS conduction capability, the MOS may enter the saturation working mode, and the power consumption will be sharp When it rises, the chip passes the patented overload protection circuit, and when it detects that the MOS is saturated, the power MOS is forcibly turned off to prevent the MOS from being damaged due to excessive power consumption.

7. Chip Overheating Regulation (OTP)

The OCP8195X integrates thermal regulation. When the IC temperature exceeds 145°C, the system adjusts the output current to gradually decrease, as shown in the figure below. Therefore, the output power In this way, the system temperature is effectively limited and the system reliability is improved.

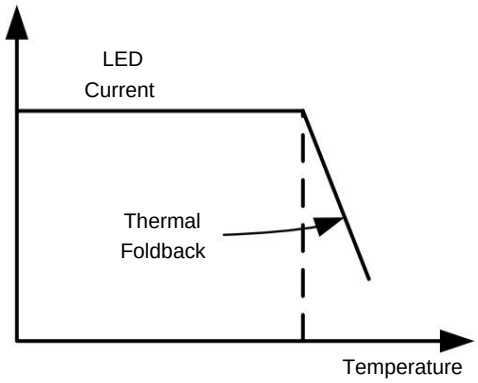


Figure 6 Schematic diagram of overheating protection

8. Night light working mode

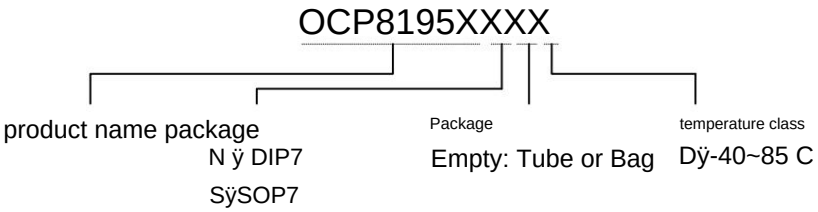
The chip integrates the night light working mode to be compatible with switches with night light function. When the night light function switch is turned off, the night light is on, and the chip detects this state After that, it works in a linear state to ensure that the night light is always on, preventing the chip from being powered on and off repeatedly and causing the night light to flicker.

9. Dimming

The chip integrated enable terminal can be floated and turned on by default. Through the external pull-down control chip, it is compatible with the dual color temperature scheme. Generate PWM signal through MCU control, can realize PWM dimming.

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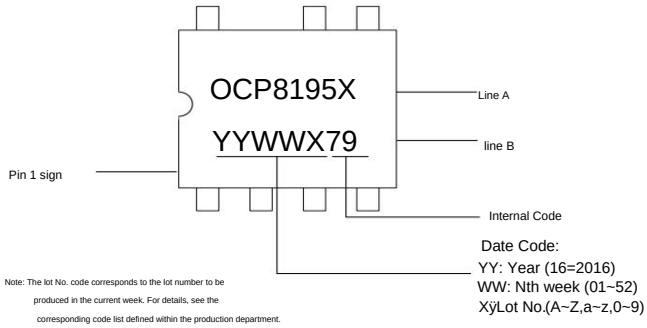
ÿ Ordering Information



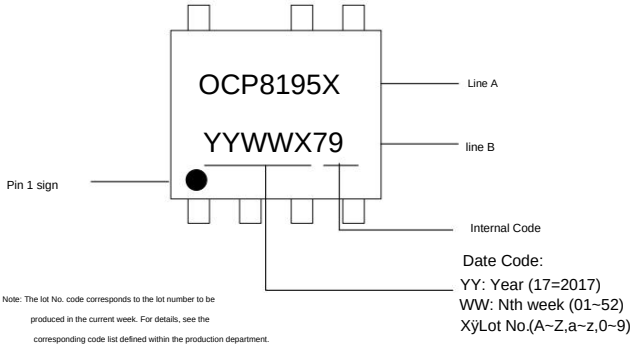
model	package	ÿÿÿÿ 13-in	temperature range	Environmental grade	base material
OCP8195CSAD	SOP-7L	reel 3500pcs/ reel 13-in reel	-40ÿ85ÿ	Green	With
OCP8195DSAD	SOP-7L	3500pcs/reel 13-in reel 3500pcs/	-40ÿ85ÿ	Green	With
OCP8195ESAD	SOP-7L	reel 13-in reel 3500pcs/reel	-40ÿ85ÿ	Green	With
OCP8195FSAD	SOP-7L		-40ÿ85ÿ	Green	With
OCP8195END	DIP-7L	Bulk 50pcs/bulk	-40ÿ85ÿ	Green	With
OCP8195FND	DIP-7L	Bulk 50pcs/bulk	-40ÿ85ÿ	Green	With

ÿ Marking information

DIP-7L



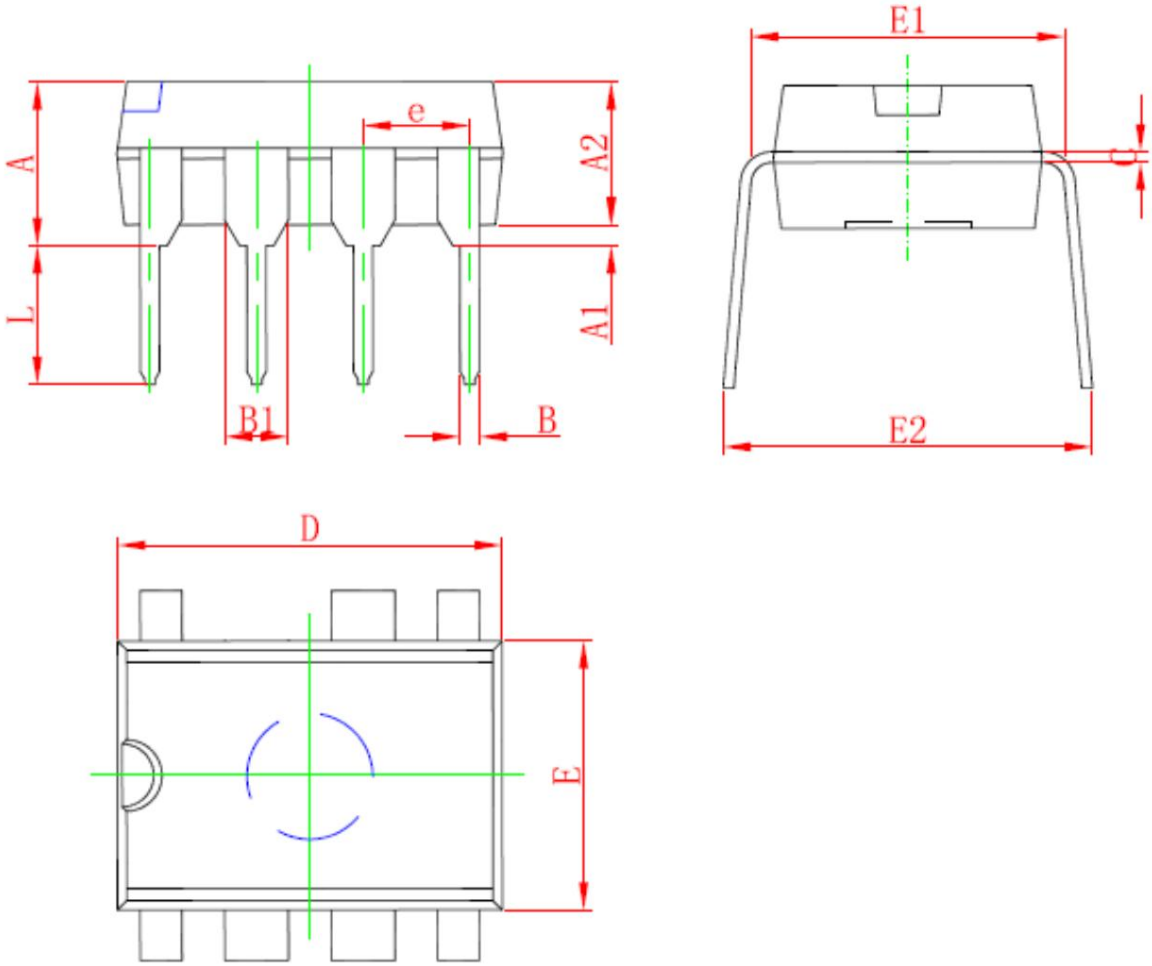
SOP-7L



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ÿ Package information

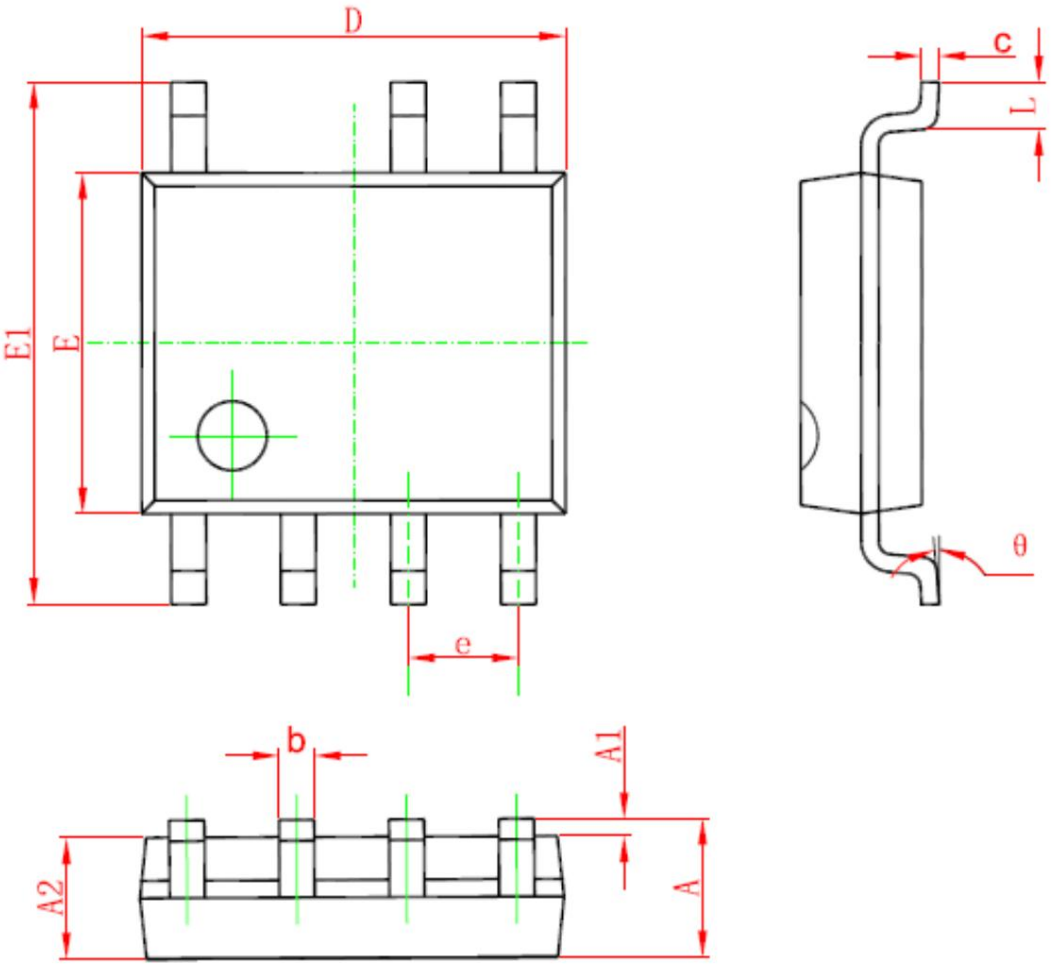
DIP-7L



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	3.710	4.310	0.146	0.170
A1	0.510		0.020	
A2	3.200 3.600 0.380 0.570		0.126 0.142	
B	1.524 (BSC) 0.204 0.360		0.015 0.022	
B1	9.400 6.200 6.600		0.060 (BSC)	
C	7.320 7.920 2.540 (BSC)	0.000	0.008 0.014	
D	3.600 8.400 9.000		0.354 0.370	
AND			0.244 0.260	
E1			0.288 0.312	
and			0.100 (BSC)	
L			0.118 0.142	
E2			0.331 0.354	

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SOP-7L



symbol	Dimensionsÿmmÿ Max		DimensionsÿInchÿ	
	Min	1.750	Min Max	
A	1.350	0.250	0.053 0.069	
A1	0.100	1.550	0.004 0.010	
A2	1.350	0.510	0.053 0.061	
b	0.330	0.250	0.013 0.020	
c	0.170	5.100	0.006 0.010	
D	4.700	4.000	0.185 0.200	
AND	3.800	6.200	0.150 0.157	
E1	5.800	1.270	0.228 0.244	
and	(BSC) 0.400 1.270 0 o		0.050 (BSC)	
L	8 o		0.016 0.050	
ÿ			0 o 8 o	

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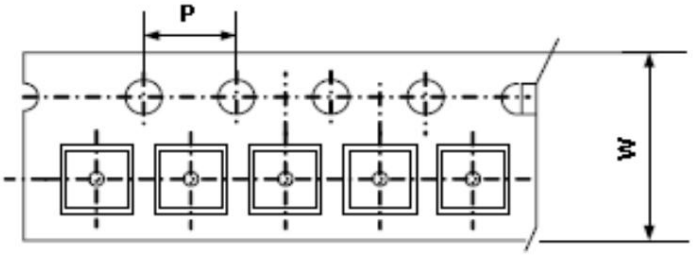
ÿ Packaging information

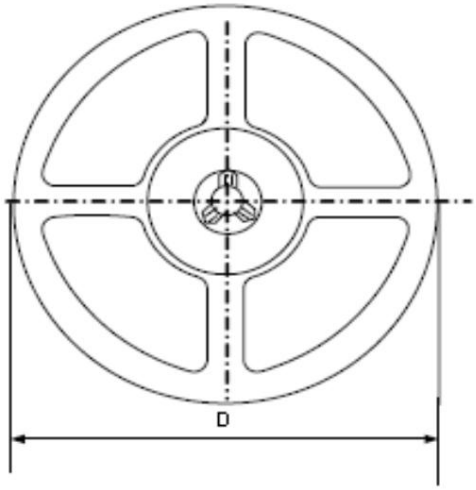
Tube **DIP-7L:**

(a), Packing Type: Tube

(b), Tube quantity: 50pcs/tube

Taping **SOP-7L:**





Package	Carrier Width(W)	Pitch(P)	Reel Size(D)	Packing Minimum
SOP-7L	12.0±0.1 mm Note:	4.0±0.1 mm	330±1 mm	3500pcs

Carrier Tape Dimension, Reel Size and Packing Minimum

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